OpenShift Deployment Guidelines and Best Practices

Deployment

Summary

The following describes guidelines and best practices, specific to the Kubernetes "Deployment" resource. This is a supplement to the OpenShift General Workload Guidelines and Best Practices.

Reference Documentation

- OpenShift General Workload Guidelines and Best Practices
- Deployment [apps/v1] - Workloads APIs | API reference | OpenShift Container Platform 4.16
- Horizontal Pod Autoscaling (HPA) Guidance and Best Practices

Deployment Best Practices

- Declare replicas greater than 0
- Declare a minimum of 3 Replicas in UAT, Perf and Prod Environments
- Declare Anti-affinity rules

- Define Horizontal Pod Autoscaling (HPA)
- Define a Pod Disruption Budget (PDB)
- Define Vertical Pod Autoscaling (VPA) in Inform mode
- Use Affinity rules to minimize cross-node traffic between dependent services
- If exposing as a webservice, label ports within the Container spec for the pod(s) (container.ports.name) and reference the port names from the Service object

Deployment Anti-Patterns

- Single Replica

Development Best Practices

- Test with a minimum of 2 replicas in lower environments
- Instances (replicas) should execute as one or more stateless processes
- Define metric(s) that can be leveraged by Horizontal Pod Autoscaling (HPA)